Project 1

Tasks:

(1) Implement a breadth-first search and a depth-first search to find a path from Arad to Bucharest in the Romania roadmap (Figure 1). Print the nodes expanded and the frontier at each step of the search process. The node should be formatted as "*Parent_city->Current_city(Path_cost)*". When the solution is found, print the solution path and the total path cost;

(2) Implement the A* search using the heuristic function in Fig. 2 with the same output requirements, except that the nodes should be formatted as "*Parent_city->Current_city(Path_cost, Estimated_total_cost)*".

(3) Extra credit: In a text cell, explain why A* is cost-optimal if the heuristic function is admissible. Read the proof on Page 88 in the textbook. Explain the proof using the specific example of planning a route from Arad to Bucharest in the Romanian roadmap with straight-line distance as the heuristic function.

Please use the provided Jupyter Notebook template for implementation and submit the completed Jupyter Notebook file that **includes code, documentation, and required output in your submission.** Provide clear comments to document your implementation. Interleaving text cells and code cells to explain your overall code structure is strongly recommended.

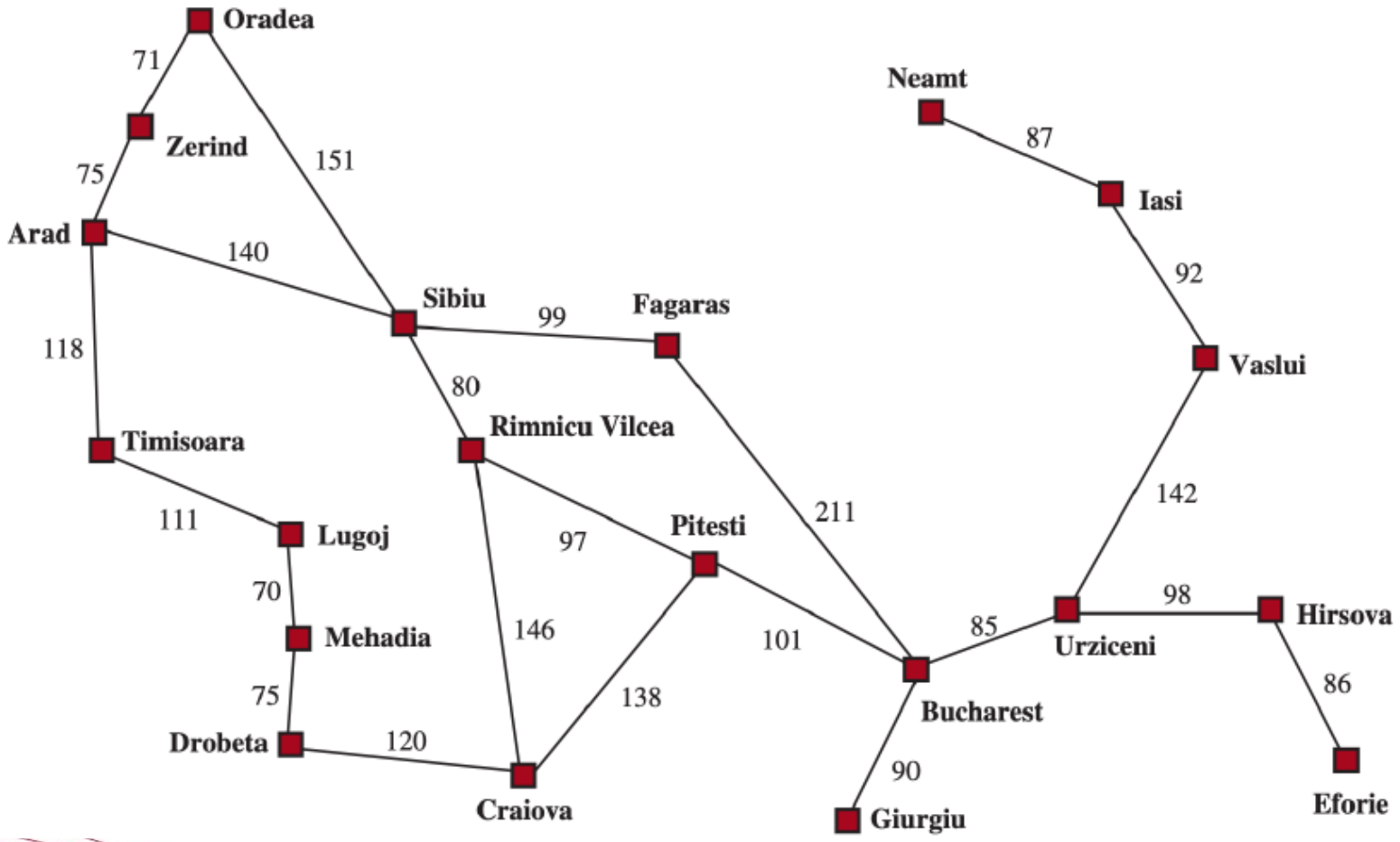

Figure 1

You may work as a group of 2 members. Please include your team member's name in your report if applicable. Otherwise, it will be treated as plagiarism if a similar code is found.

You can use utility packages, but the core search algorithm must be implemented by you.

Grading:

- 40% for the successful execution and correct output from your BFS and DFS implementation
- 40% for the successful execution and correct output for A* search implementation
- 20% for documentation, code structure, and readability
- Up to 20% for the extra credit task

CHAPMAN UNIVERSITY

Arad	366	**Mehadia**	241
Bucharest	0	**Neamt**	234
Craiova	160	**Oradea**	380
Drobeta	242	**Pitesti**	100
Eforie	161	**Rimnicu Vilcea**	193
Fagaras	176	**Sibiu**	253
Giurgiu	77	**Timisoara**	329
Hirsova	151	**Urziceni**	80
Iasi	226	**Vaslui**	199
Lugoj	244	**Zerind**	374

Figure 2. Values of h_{SLD}—straight-line distances to Bucharest.